

Control Relays

Lecture 12

Control Relays

- A control relay is an electrically operated switch that uses a small amount of power to control a separate, larger electrical circuit.
- It contains a conducting coil that, when energized by a low-power signal, creates a magnetic field to move contacts, which either opens or closes a switch in the higher-power circuit. This allows a low-power signal to remotely and safely control high-voltage or high-current loads, a function essential for automation and circuit protection.
- Relays are magnetic electromechanical devices with two primary purposes: to isolate different circuit voltages, and to form larger complex networks of logic to run machines without digital controllers.



An electromechanical relay is an electrical switch actuated by an electromagnet coil. As switching devices, they exhibit simple “on” and “off” behavior with no intermediate states. Relays are very useful devices, as they allow a single discrete (on/off) electrical signal to control much greater levels of electrical power, and/or multiple power or control signals that are otherwise isolated from each other. For example, a relay may be controlled by a low-voltage, low-current signal that passes through a delicate switch of some sort (e.g. limit switch, proximity switch, optical sensor), and then the switching contacts of that relay may be used to control a much higher-voltage, higher-current circuit, and even multiple circuits given multiple sets of switching contacts. *Single-pole Single-throw Relay (SPST Relay)*

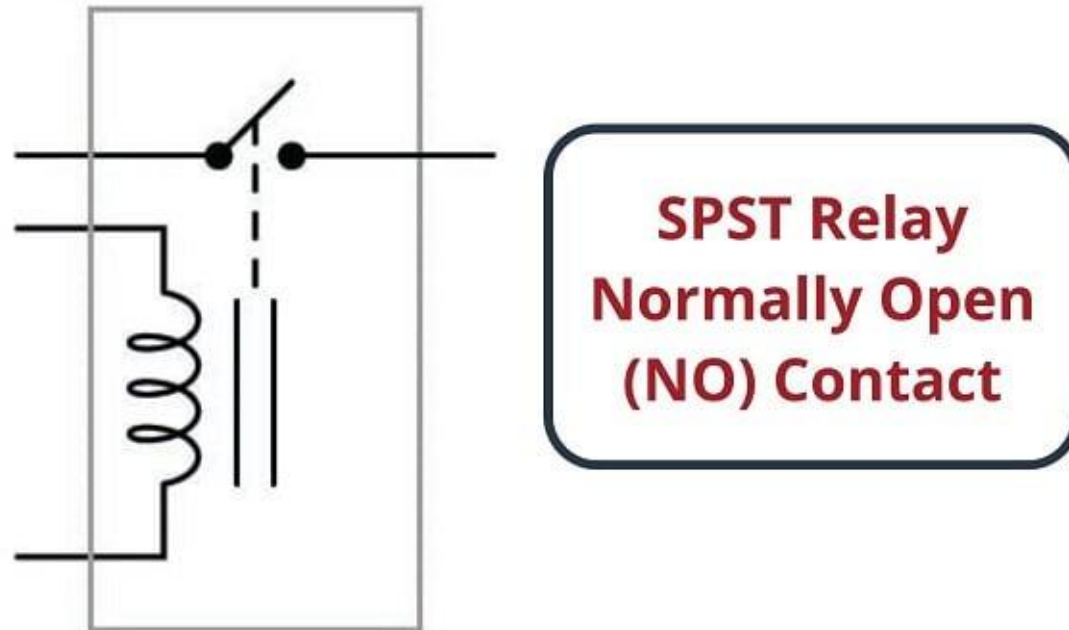
Control Relay components

All **control relays** come with the following components:

- Mechanically movable contact
- Electromagnet coil
- Spring
- Switching points

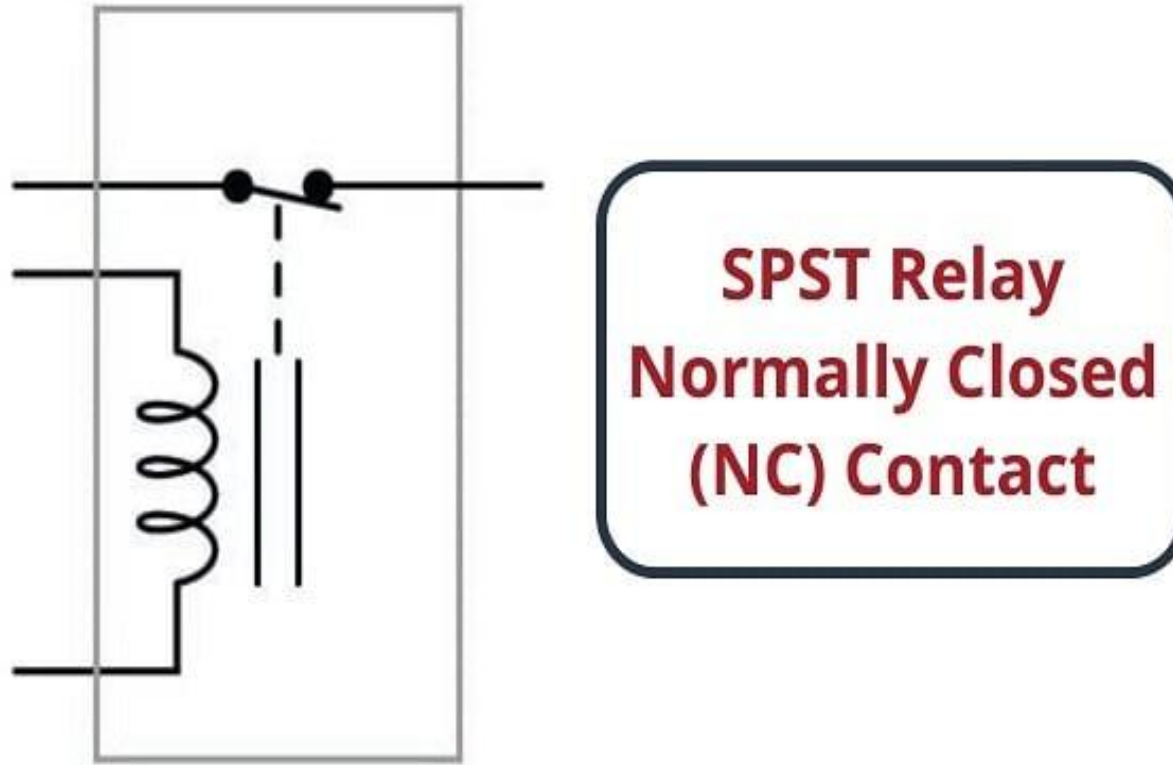
Schematic Symbols for Relays

- ***Single-pole Single-throw Relay (SPST Relay)***
- The electronic schematic symbol for a simple single-pole, single-throw (SPST) relay is shown here:

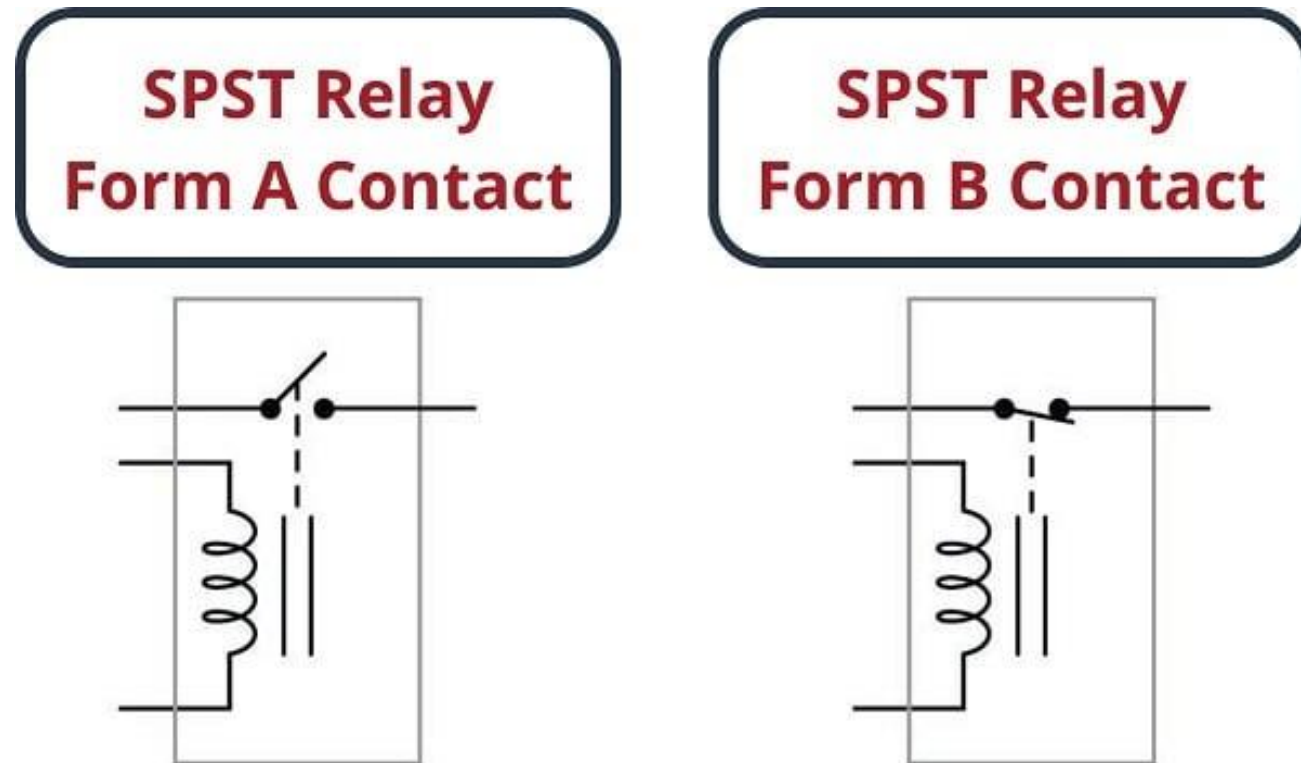


A coil of wire wrapped around a laminated ferrous core provides the magnetic field necessary to actuate the switch mechanism, energized by [either AC or DC voltage](#), depending on the model. This electromagnet coil's actuating influence on the relay's contact(s) is represented by the dashed line. This particular relay is equipped with *normally open* (NO) switch contacts, which means the switch will be in the open (off) state when the relay coil is de-energized. Recall that the “normal” status of a switch is the resting condition of *no stimulation*. A relay switch contact will be in its “normal” status when its coil is not energized.

A single-pole, single-throw relay with a normally-closed (NC) switch contact would be represented in an electronic schematic in the following Figure:

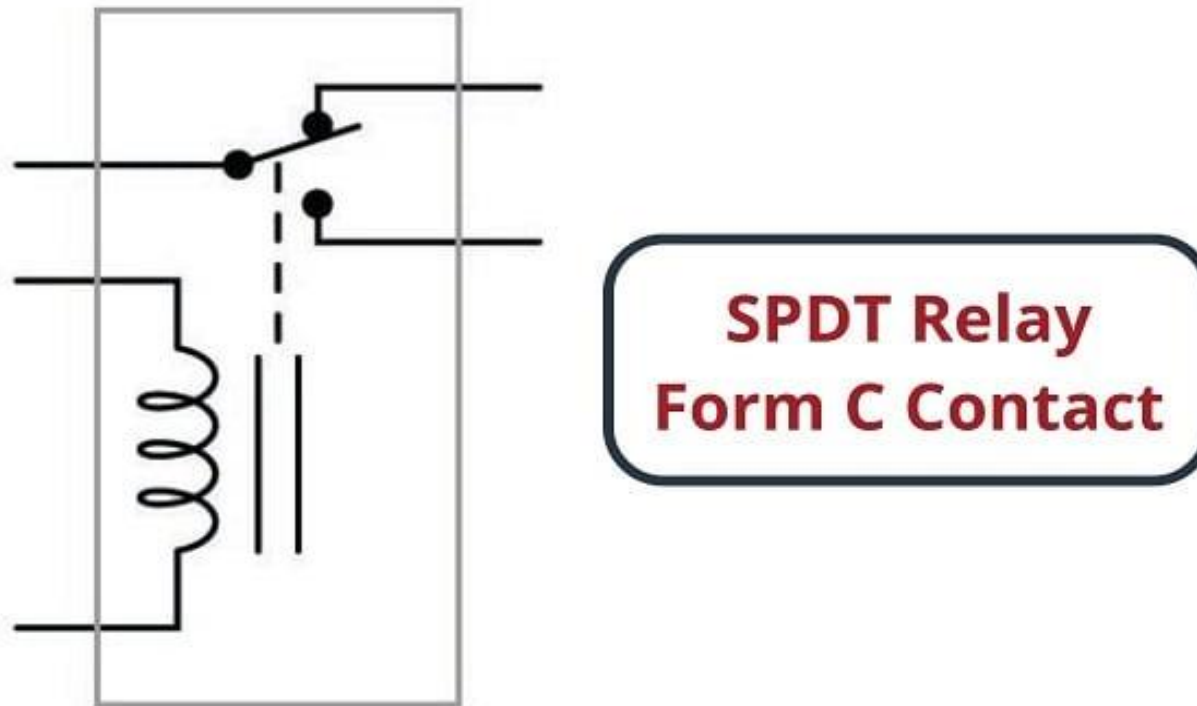


In the electrical control world, the labels “Form-A” and “Form-B” are synonymous with “normally open” and “normally closed” contacts, respectively. Thus, we could have labeled the SPST relay contacts as “Form-A” and “Form-B,” respectively



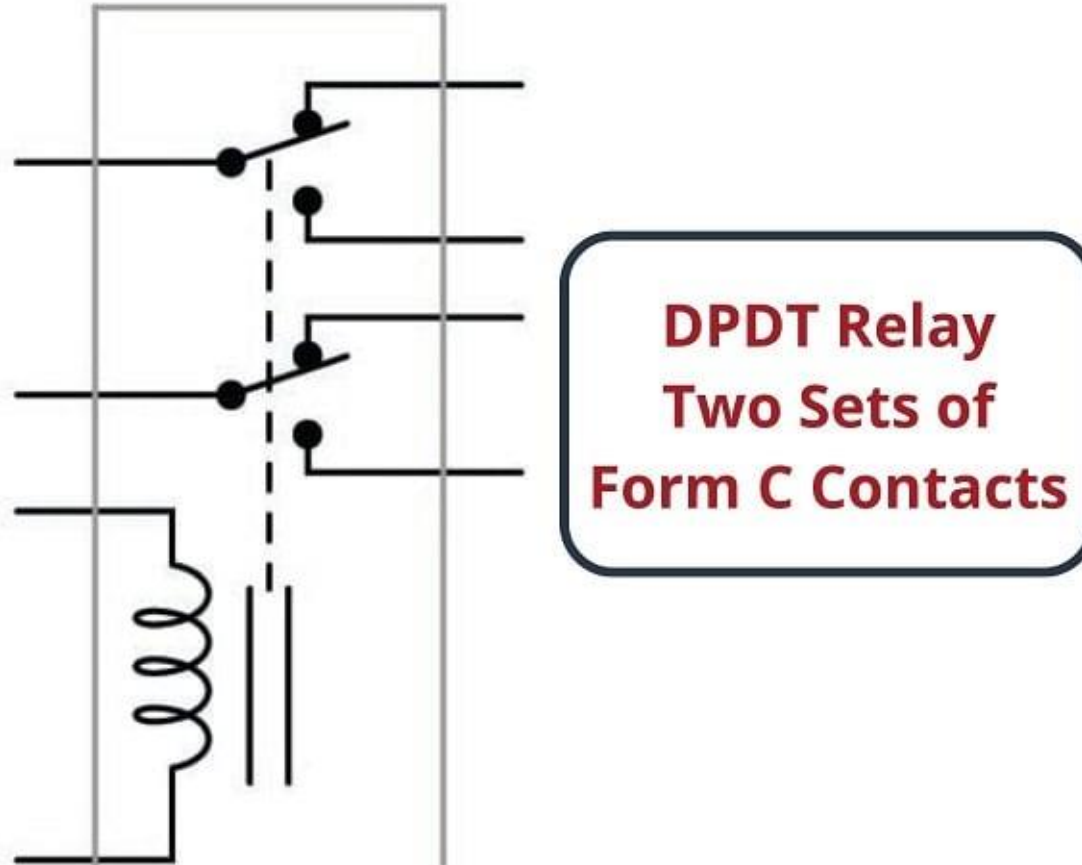
Single-pole Double-throw Relay (SPDT Relay)

The single-pole, double-throw (SPDT) relay contact, known as a “Form-C” contact. This design of switch provides both a normally-open and normally-closed contact set in one unit, actuated by the electromagnet coil:



Double-pole Double-throw Relay (DPDT Relay):

The double-pole, double-throw (DPDT) relay contact, provides two sets of Form-C contacts in one unit, simultaneously actuated by the electromagnet coil:



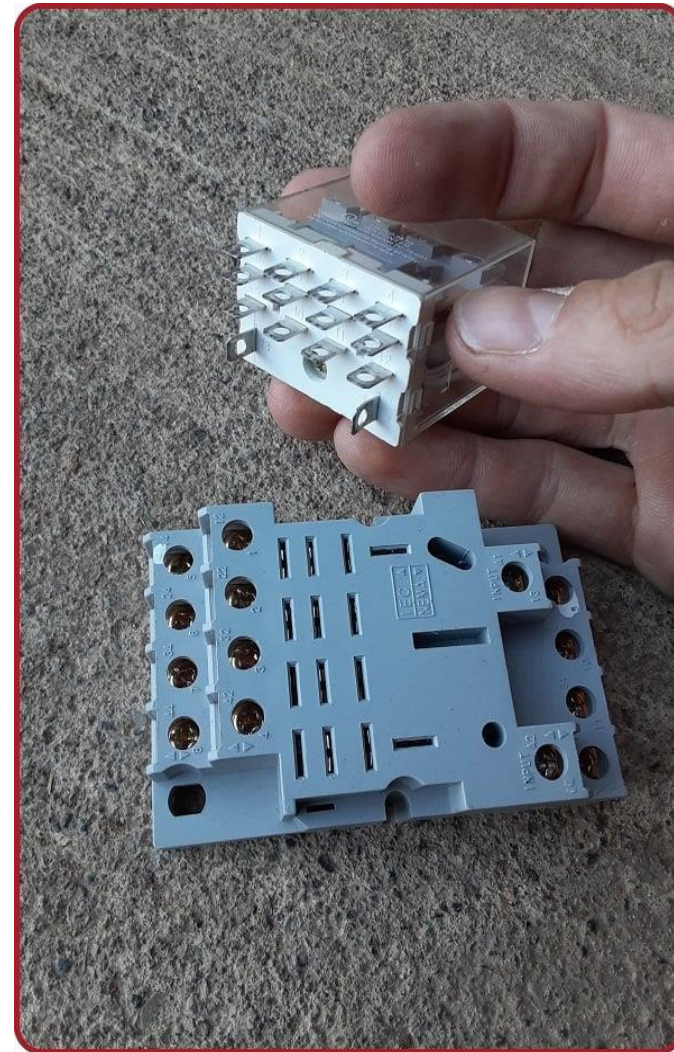
DPDT relays are some of the most common relay form types found in industry due to their versatility. Each Form-C contact set offers a choice of either normally-open or normally-closed contacts, and the two sets (two “poles”) are electrically isolated from each other so they may be used in different circuits.

Two common packages for industrial relays are the so-called [octal relay and the ice cube relay](#). These relays plug into multi-pin base sockets for easy removal and replacement in case of failure, and typically contain clear plastic cases allowing inspection of the working elements.

The “ice cube” relay is functionally identical, but the pins are arranged in a rectangular pattern, rather than circular around the base. These relays can be found in single and double-pole varieties, and even up to 4 contact sets (4PDT) is fairly common.



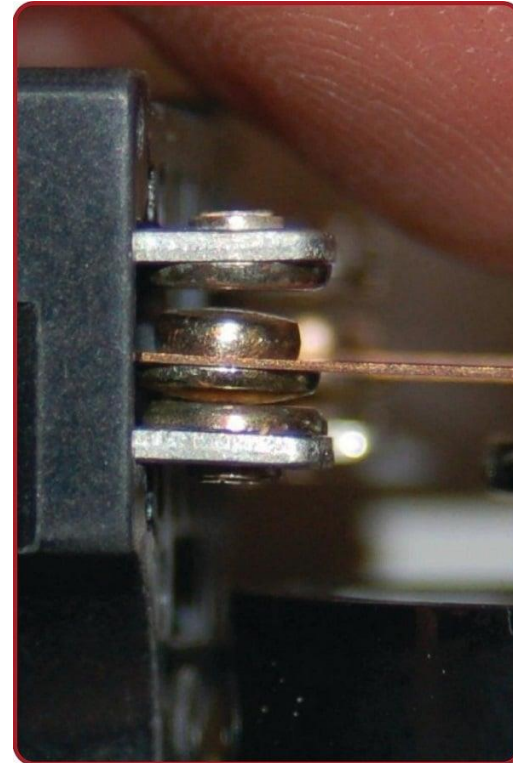
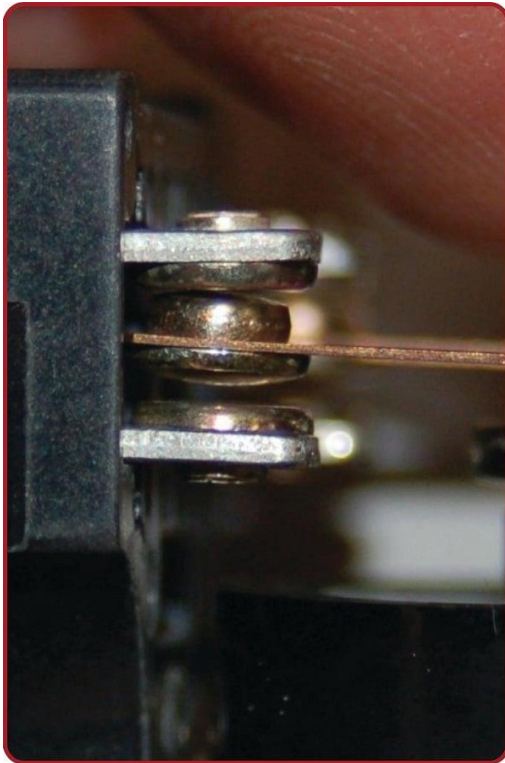
Octal Relay



Ice Cube Relay

These relays connect to the socket with eight pins: three for each of the two Form-C contact set, plus two more pins for the coil connections. Due to the pin count (8), this style of relay base is usually reserved for DPDT connection style.

A closer view of one Form-C contact shows how the moving metal “leaf” contacts one of two stationary points, the actual point of contact being made by a silver-coated “button” at the end of the leaf. The following photographs show one Form-C contact in both positions:



A Control Relay Work Principles

- **Energizing the coil:** A small current is sent through the relay's internal coil.
- **Creating a magnetic field:** The current flow creates a magnetic field.
- **Activating the switch:** The magnetic field pulls a switch mechanism, which can either close a normally open contact or open a normally closed contact in a separate circuit.
- **De-energizing the relay:** When the current to the coil is turned off, the magnetic field disappears, and the switch returns to its original position.

Control relays applications

- **Automation:** They are critical for automating processes in industrial systems, computer interfaces, and other electronic devices.
- **Safety:** Control relays provide isolation between low-power control circuits and high-power circuits, protecting equipment and personnel from high voltages.
- **Circuit control:** They allow a single low-power signal to control multiple circuits or to switch circuits on and off remotely.
- **Protection:** Some specialized control relays, like overload or earth-fault relays, are designed to detect and protect against electrical faults.
- **Flexibility:** They are used in a wide range of applications, from simple lighting systems to complex motor controls and automotive electronics.

Control Relays Types

- Different types of control relays are available to suit specific application needs:
- **Electromechanical Relays (EMRs):** These are traditional relays that use a physical electromagnet and mechanical contacts to open or close a circuit. They are widely used due to their simplicity and cost-effectiveness.
- **Solid State Relays (SSRs):** These relays use semiconductor components for switching, without any moving parts. They are favored for applications requiring high-speed switching, long life (billions of operations), and resistance to vibration and shock.

Thermal Overload Relays: Designed to protect motors and other equipment from overheating or damage due to prolonged excessive current flow. They function on the principle of thermal sensing, tripping when a bimetallic strip heats up and deforms.

Latching Relays: These relays maintain their contact position even after the control signal is removed, using a permanent magnet or mechanical latching mechanism. A second coil or pulse is typically used to reset the relay to its original state.

Time-Delay Relays: These incorporate a timing function to delay the opening or closing of contacts, which is useful in sequential operations where timing is critical.

Reed Relays: Consist of two magnetic strips (reeds) sealed inside a glass tube, known for precision and reliability in low-current applications

Control relays Specifications

- When selecting a control relay, consider the following specifications:

Specification	Description
Coil Voltage	The voltage required to energize the relay coil (e.g., 5V, 12V, 24V DC or 120V, 240V AC).
Contact Rating	The maximum voltage and current the contacts can safely switch (e.g., 10A at 120VAC, 5A at 30VDC).
Contact Form/ Configuration	Describes the number of poles and throws, such as Single Pole Single Throw (SPST), Single Pole Double Throw (SPDT), or Double Pole Double Throw (DPDT).
Mounting Type	How the relay is installed (e.g., DIN rail mount, panel mount, PCB mount, plug-in).
Operating Temperature	The environmental temperature range in which the relay is designed to function reliably.

Control relays Applications

- Control relays are used across a wide variety of applications:
- **Industrial Automation:** Controlling motors, solenoids, and heavy machinery in factories and power plants.
- **Automotive Industry:** Managing engine systems, lighting, and entertainment systems.
- **Building Automation:** Regulating HVAC systems, lighting, and security features in commercial and residential buildings.
- **Telecommunications:** Managing data transmission and controlling circuits remotely.
- **Safety and Protection:** Providing overload protection and fault isolation in electrical power systems.